

at least four to five minutes to generate a post-reaction colour. Users have to conduct multiple tests to find out the right adulterant.

Outcomes of the tests are also difficult to pinpoint as colour changes are often hard to distinguish. There is also an environmental constraint when using chemicals.

In the market, a strip test facility is also available, which changes colour if dipped in adulterated milk for about five minutes. However, the strips are expensive and not practical for real-life situations, like in rural areas where there are long queues for milk collection, or in households where packaged milk is directly used.

Later, electronic devices were imported for dairy testing. But these had a big form factor, and each unit's cost was ₹ 10 million or above, with maintenance costs of ₹ 5000 to ₹ 10,000 every three to four months. Sensors malfunctioned due to lack of cleaning. Looking at all these constraints, CEERI got to work to find a compact and cost-effective alternative.

Benefits of the portable tester

In an initial project of developing automated tea quality test technology, the team at CEERI designed a system mimicking a human being testing tea—through colour, smell and taste. Starting from that project, the team developed a system using a computer vision-enabled camera for colour identification, an electronic nose for

odour identification and, finally, an electronic tongue to test taste. The design prompted the application of these technologies further into milk and other beverages like tea, juice and wine.

The system works on a simple electrochemical principle. A metal alloy electrode, when dipped in a liquid, measures the electrochemical signature of the sample. Pure milk has an electrochemical signature that changes with the introduction of adulterants. The electrode detects if there is a change in the electrochemical signature due to reaction by ions of the adulterant.

This allows users to know if the milk is adulterated. The system can be Wi-Fi-enabled with cloud connectivity, to store data remotely. It may also be connected to a smartphone through cables. A 9V lithium-ion battery powers the system.

This technology can be highly beneficial for dairy businesses to ensure quality of milk they are selling. Usually, milk is procured from a collection centre and is sent to a chilling centre for preservation at proper temperature. From there, it reaches the dairy outlet in the market.

Having a device at all locations can help businesses stay informed about the quantity, quality and preservation of milk at each stage. Whenever any deviation is noted at any location, remedial actions can be taken immediately. This way, outlet owners can

stop adulterated milk from reaching the market.

Alternatively, the technology developed by CEERI is economical enough to be purchased by end consumers, who can test a sample of the milk they purchase before consumption.

Commercial use

The team at CEERI estimates that the bill of material of the technology will be around ₹ 5000. So far, it has collaborated with Rajasthan Electronics Instruments in Jaipur and some other product developers, whom they are transferring the technology to, for manufacturing the product.

CEERI provides one year of hand-holding, starting with providing design documents, hex files, programs, as well as guidance in terms of manufacturing, test and debugging of the first 25 products being manufactured, all without any charge.

CEERI researchers recall miniaturising the device, formulating the circuit design, component selection and fitting the program in an 8-bit controller as challenging tasks.

The system, once commercialised, has the potential to save consumers from adulterated dairy-driven health issues. At the same time, it can enable dairy providers to be more cautious about their product quality, thus saving their market reputation, as well as major expenses from wastage by enabling quick root-cause analysis.

4

GREEN TECH:

“Turning To **EV REDUCES** The Overall **BILL** Of Material By **30 TO 40 PER CENT**”

- India is making quick strides towards the promised electric vehicle (EV)-driven transportation ecosystem. According to a report by TechSci Research, India enjoyed a US\$ 27 million two-wheeler EV market in 2017, which is poised to grow to US\$ 200 million by 2023, at a CAGR of 41 per cent. Considering three-wheelers, passenger cars and electric buses in the mix, overall CAGR will continue to grow at 37 per cent. However, of late, four-wheelers have not seen much deployment compared to two-wheelers in India. Paromik Chakraborty of *Electronics For You* caught up with Jeetender Sharma, managing director, Okinawa Autotech Pvt Ltd to understand the prospects of EVs in India